SOP-IT-010 · How we keep our data: retention, backups, archives

SOP IT-010 | Status draft | Changed 2026-11-06 | For Anyone who keeps work on their own Mac

Department: IT · **ID:** 010 · **Status:** draft (approved when tagged) · **Owner:** the lab; any person may run the checks · **Last changed:** 2026-11-06

> ***Note. TL;DR.*** *Each kind of data has one place that is its record, and at least one copy somewhere else.*
>
> - ***Transcripts:*** *Claude Code deletes them after **30 days** unless told otherwise, silently. We set `cleanupPeriodDays` to **3650** (ten years), and seal a copy of `~/.claude/projects` every night, encrypted, into iCloud Drive or a folder you name. The live `~/.claude` never goes into iCloud.*
> - ***Code:*** *git and GitHub are the record. The safety bundles carry what GitHub does not have yet.*
> - ***The whole Mac:*** *Time Machine, once a destination exists. Leave out only what can be rebuilt.*
> - ***Databases are readings, raw logs are the record.*** *Back up the logs; rebuild the database.*
> - ***Never to GitHub:*** *transcripts, keys, passphrases, `.env` files, databases, archives. **Passphrases and keys live in the password manager.***
> - ***A copy you have never restored from is a hope.*** *Run the restore drills on their schedule.*

If you want to know	Go to
Why old sessions vanish, and how to keep them	1. Transcript retention
How the nightly sealed copy works, and how to get it back	2. The nightly archive
What Time Machine should leave out	3. Time Machine
What protects uncommitted work	4. Git and the safety bundles
Which copy is the record	5. Readings and records
What must never be pushed	6. What never goes to GitHub
How often to prove it works	7. The restore drills

Purpose

Say, in one place, how we keep each kind of data: how long it is kept, where its record lives, where its copies go, and how we prove we can get it back. The procedures are short and each has commands, so anyone can run them and anyone can check.

Scope

A Mac where a person works with Claude Code and git: their transcripts and agent memory, their repositories, the logs and databases a job writes, and the Mac as a whole. **Not covered:** wallets and seeds (SOP-IT-009), encrypted disk images (SOP-IT-008), FileVault and the other guards (SOP-IT-007), and which drive or plan to buy.

Roles

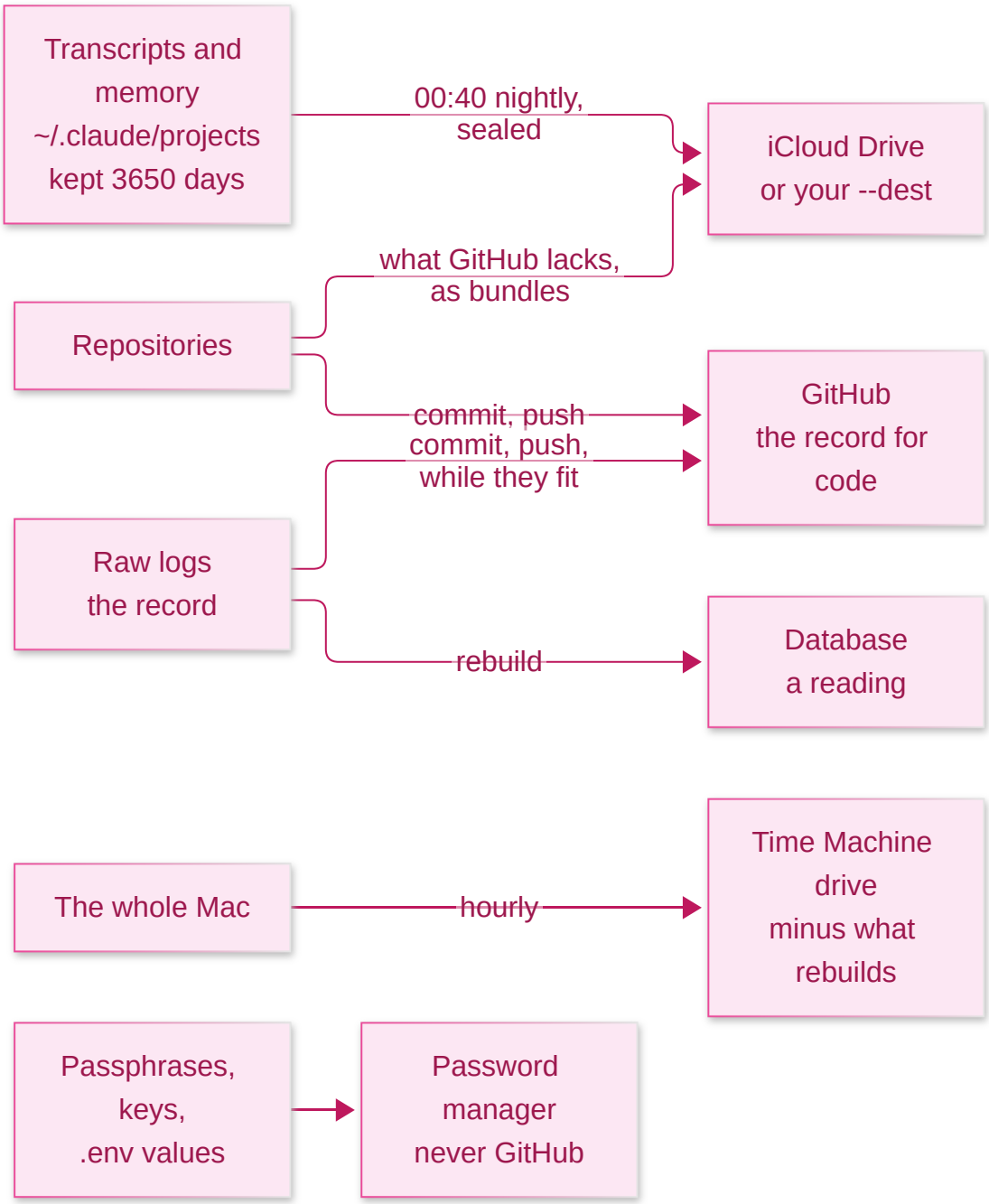
Role	Does
The person at the keyboard	owns the Mac and its data; runs every step that changes their settings, makes a passphrase, schedules a job, or may put a prompt on the screen; keeps the passphrases in the password manager

A steward	keeps this page true; checks the drill log each quarter
Helper (an agent)	may run the self-tests, dry runs, `--list` and `--verify`, and hand over the other lines as `!` commands. **Never** runs `--init`, `--install` or the settings installer, and never reads or prints a passphrase, a key, a `.env` file or the values in `~/.claude/settings.json`

Prerequisites

- A clone of `my-template` at `~/code/my-template` (the scripts below live in its `scripts/` folder).
- A password manager the person already uses.
- For the archive's default destination: iCloud Drive turned on. Without it, a folder or drive of your own (`--dest`).

How it works



Data	Its record	Its copies	Kept
Transcripts, agent memory	`~/.claude/projects` on this Mac	the nightly sealed archive; Time Machine	ten years on the Mac; archives are never deleted
Code	GitHub, once pushed	a fresh clone; the safety bundles for what is not pushed; Time Machine	for as long as the repository lives
Raw logs	git, while they fit; an off-machine copy once they leave it	Time Machine	as their procedure says
Databases built from logs	none: they are readings	none needed: rebuild from the logs	until rebuilt

Passphrases, keys, `.env` values	the password manager	the password manager's own backup	as long as what they open

1. Transcript retention

Why. Claude Code keeps session transcripts for **30 days by default** and then deletes them in a background sweep after a session starts, **without a message**. A session older than that drops out of `/resume`, and its history is gone. The setting is `cleanupPeriodDays`: a whole number of days, at least 1 (Claude Code's settings reference, read Nov 6, 2026). We set it to **3650**, ten years.

Set it, once per Mac. The person runs the installer themselves; a helper must not, because it changes their own settings and shell profile.

1. Run `sh ~/code/my-template/scripts/install-claude-session-order.sh`. It sets `"cleanupPeriodDays": 3650` (or keeps a larger value), adds the session-start hook and the parent-folder guard, keeps every other key as it was, and checks the file is valid JSON before it replaces the old one. *You should see:* the list of key names (never their values), then `cleanupPeriodDays: 3650`, and `Done.`
2. **Its backups, made on the first install only:** `~/.claude/backups/settings-before-session-order.json` and `~/.zshrc.before-session-order`. They are kept if they already exist, so they always hold the state from before the first install. Running the installer twice is safe.
3. **To undo:** `sh ~/code/my-template/scripts/install-claude-session-order.sh --undo` puts both files back from those backups. *You should see:* `restored` and each file's path.

Check it, reading only that one key. `settings.json` can hold tokens, so never print the whole file:

```
python3 -I -c 'import json,os;
print(json.load(open(os.path.expanduser("~/.claude/settings.json"))).get("cleanupPeriodDays", "not
set: 30 days"))'
```

You should see: `3650`. A project's own `.claude/settings.json` or `.claude/settings.local.json` outranks your user file in that project, so look there too. This prints only the lines naming the key:

```
find ~/code ~/dappu ~/drasticstaticLabs -maxdepth 3 -path '*/.claude/settings*.json' -exec grep -Hn
'"cleanupPeriodDays"' {} + 2>/dev/null
```

You should see: nothing, or only values of 3650 or more.

2. The nightly archive of `~/.claude/projects`

Why. Keeping transcripts for ten years keeps them on one disk. `~/.claude/projects` holds every session and every agent's memory, is in no git repository, and must never go to GitHub. So a script seals a copy each night: `scripts/claude-projects-archive.sh`. The walkthrough is `my-template/walkthroughs/CLAUDE-PROJECTS-ARCHIVE.md`.

What it does.

- Archives everything under `~/.claude/projects`, and `~/.claude/CLAUDE.md`. Nothing is left out. The rest of `~/.claude` (settings, credentials, caches) stays out.
- `tar`, `gzip`, then `openssl enc -aes-256-cbc -pbkdf2 -iter 600000`. The passphrase is read from `~/.config/claude-archive/passphrase` (mode 600), never typed on a command line.
- **A full archive** on the first run for a destination, on the first run of each month, and with `--full`. Otherwise **an incremental**: the files changed since the last success. A transcript that grew by one line travels whole.

- Names carry the date and time, `claude-projects_YYYYMMDD-HHMMSS_full.tar.gz.enc` or `..._incr.tar.gz.enc`, each with a `.sha256` beside it. A plain sort of the names is the restore order.
- Each archive is written as `.partial`, decrypted again and counted, then renamed. A failed run leaves nothing and is retried. **No archive is ever deleted.**
- Default destination: `~/Library/Mobile Documents/com~apple~CloudDocs/Backups/claude-projects/YYYY/` (iCloud Drive).
- The nightly job is a LaunchAgent at **00:40** that runs `--if-due`: it also runs at login after a night the Mac was off, and does nothing if the last success is under 20 hours old.

Measured Nov 6, 2026 into a scratch folder, not iCloud: a full archive of 1,913 files went from 2.8 GB to **1.2 GB** in 64 s, plus 7 s to verify; one day's incremental, 285 files, 557 MB to **216 MB** in 12 s. At that pace it is roughly **4 to 8 GB a month, 50 to 100 GB a year**, and growing. Check your iCloud plan before a year has passed.

Never put the live `~/.claude` in iCloud Drive, and never link it there. iCloud syncs files one at a time while Claude Code is still appending to them, so the cloud can hold half a transcript, or a conflicted second copy. "Optimize Mac Storage" can move a file off the disk, leaving a placeholder Claude Code cannot read. And `~/.claude` holds settings and tokens that do not belong in any sync. Only finished, sealed archives travel. The same goes for `~/Desktop` and `~/Documents` when iCloud's *Desktop & Documents Folders* is on: they are then iCloud Drive too.

Set it up. The person at the keyboard runs steps 2 to 5; a helper may run step 1 and hand over the rest.

1. **Prove the script on this Mac:** `sh ~/code/my-template/scripts/claude-projects-archive.selftest.sh`. It works in a temporary folder with a fake home. *You should see:* every check pass (138 of 138 on Nov 6, 2026).
2. **Make the passphrase:** `sh ~/code/my-template/scripts/claude-projects-archive.sh --init`. It writes a random passphrase only if there is none, and never shows it. *You should see:* the folder made and a `pbcopy` line to run next.
3. **Put it in the password manager,** then clear the clipboard. Make a password item (for example *claude-projects archive passphrase*) and paste between these two lines:

   ```
   pbcopy < ~/.config/claude-archive/passphrase
   pbcopy < /dev/null
   ```

   You should see: the item saved. Without this copy, a lost Mac means every archive is noise.
4. **The first archive, by hand:** `sh ~/code/my-template/scripts/claude-projects-archive.sh --run`. **This may put a prompt on your screen:** the first time Terminal writes into iCloud Drive, macOS may ask for access. *You should see:* a full archive, its size and the destination's total. iCloud uploads it in the background.
5. **Schedule it:** `sh ~/code/my-template/scripts/claude-projects-archive.sh --install`. macOS shows a *Background Items Added* notification; that is expected. *You should see:* the agent loaded. To turn it off later: `--uninstall` (the plist is moved aside, not deleted, and the archives stay).
6. **The morning after, once:** `sh ~/code/my-template/scripts/claude-projects-archive.sh --status`. *You should see:* `ok: the newest success is under 36 h old`. A failed night is never silent: it is recorded, shows a macOS banner, and goes to `~/.config/claude-archive/notify` if that file is executable (a one-line push to your phone). Where a nightly desk job exists, have it run `--status` and alert when it exits 1. If the log says *Operation not permitted*, macOS kept the background job out of iCloud Drive: schedule a folder outside iCloud instead (`--install --dest ~/Archives/claude-projects`, which Time Machine copies) and make the iCloud copy by hand from Terminal once a week. Do not give `/bin/sh` Full Disk Access.

No iCloud, or a drive as well. Any mode takes `--dest DIR`, or `CLAUDE_ARCHIVE_DEST=DIR`:

```
sh ~/code/my-template/scripts/claude-projects-archive.sh --run --dest /Volumes/Backup/claude-projects
sh ~/code/my-template/scripts/claude-projects-archive.sh --install --dest /Volumes/Backup/claude-projects
```

The folder must already exist, so an unplugged drive cannot quietly become a folder on the startup disk: the run is refused and the next night tries again. Each destination keeps its own chain, starting with a full, and its own nightly schedule: `--install --dest DIR` adds a LaunchAgent beside iCloud's rather than replacing it (`--uninstall --dest DIR` removes only that one). A destination inside `~/.claude` is refused.

Get it back. Always into a new, empty folder in your home, never over the live `~/.claude` (the script refuses that):

```
sh ~/code/my-template/scripts/claude-projects-archive.sh --restore-latest ~/claude-restore
diff -rq ~/claude-restore/.claude/projects ~/.claude/projects
```

You should see: the newest full and every incremental after it unpacked in order, then only the differences since last night. To copy anything back, quit every Claude Code session first and copy only what you need. For an earlier day, `--restore FULL INCR ... DIR` with the files oldest first. On a new Mac, put the passphrase file back from the password manager first; the walkthrough has the three lines.

Keeping it smaller. `--prune-plan 3` prints what keeping only the newest three full archives would move aside; it changes nothing. `--move-old 3 DIR` then moves those older chains (never deletes them) to a folder such as the backup drive, each checked against its `.sha256` . Suggested rhythm: three months in iCloud, the rest moved monthly. Check the iCloud plan (System Settings → your name → iCloud) **before** scheduling: a full iCloud also stops iPhone backups and photo uploads.

3. Time Machine

Why. Time Machine copies the whole Mac to a drive you own, hourly, without the internet. It is the only copy of anything that is in neither git nor the archive: gitignored logs, local databases, app settings. It starts once a destination exists.

> ***Warning.*** *Some `tmutil` commands may mount the backup disk or ask for Full Disk Access. Tell the person at the keyboard before running one, and let them run it.*

1. **Is there a destination?** `tmutil destinationinfo` . *You should see:* a destination's name. If it says there are none, the rest of this part waits for a drive: a USB or Thunderbolt drive about twice the size of the Mac's disk, used only for backups, or a network share that speaks SMB.
2. **Add it:** System Settings › General › Time Machine › Add Backup Disk. Let the first backup finish.
3. **Leave out only what rebuilds.** A database filled from logs, `node_modules` , `.venv` , build output and caches come back by running a command; they change often and make every hourly backup larger. Mark each one; the mark travels with the item:

   ```
   tmutil addexclusion path/to/the-rebuildable.sqlite
   tmutil isexcluded path/to/the-rebuildable.sqlite
   ```

   You should see: `[Excluded]` . **Do not exclude** `~/.claude` , raw logs, or repositories. Virtual machines are the one larger exception: their makers advise copying a shut-down VM's folder to the drive yourself, outside Time Machine. The mark can be set **now, before any drive exists**: it is a flag on the item itself, needs no administrator password and shows no prompt. To take it off: `tmutil removeexclusion path/to/it` . Example, a desk database rebuilt nightly from its logs: `tmutil addexclusion ~/path/to/the-repo/logs/db` .
4. **Check it ran:** `tmutil latestbackup` . *You should see:* a date from the last hour or so.

Out of sight is not out of the backup. Hiding a folder from Finder is a different tool, for folders that confuse more than they help (one an app keeps re-creating, for example) but must stay where they are:

```
chflags hidden ~/SomeFolder      # gone from Finder; still on disk, still in Time Machine and in
Terminal's ls
```

```
ls -lO ~ | grep SomeFolder       # the word "hidden" in the flags column
chflags nohidden ~/SomeFolder    # back in Finder
```

In Finder, **Command-Shift-Period** shows or hides every hidden item for a moment. Hiding is not privacy (anyone with the account can still open it) and not an exclusion: Time Machine still copies a hidden folder, and `tmutil addexclusion` is what leaves a folder out. Use neither to "delete" something: move it into an archive folder instead.

4. Git and the safety bundles

Why. GitHub keeps what is committed and pushed. Everything else in a clone, the staged and unstaged edits, untracked files, stashes, branches not pushed, lives on one disk until it is pushed. `scripts/git-safety-backup.sh` puts exactly that into iCloud Drive as finished, verified git bundles. Its walkthrough is `my-template/walkthroughs/ICLOUD-AND-GITHUB-TOGETHER.md`.

The rules. Commit often and push the working branch; a push is the real backup. Never keep a repository in iCloud Drive: it syncs files one at a time, and a repository only makes sense whole.

What the bundles carry, and leave out. Each repository is backed up once, from its main worktree, with every linked worktree's uncommitted work. Only what GitHub lacks goes in, so bundles are small. Paths named like secrets, ignored files, `node_modules` and your exclude list stay out. A repository mid-rebase or mid-merge is left until the next run. The newest seven bundles per repository are kept, under `GitBackups/` in iCloud Drive (or `--dest`).

1. **Once per Mac, the person runs** `sh ~/code/my-template/scripts/git-safety-backup.sh --init`. It writes the skip list, the exclude list and the passphrase file, and shows the passphrase **once**: put it in the password manager. Nothing runs until the skip list exists, so a private repository is never sent before someone has thought about it. *You should see:* the three files' paths. Read the skip list and add any repository that must never leave the Mac.
2. **Dry run:** `sh ~/code/my-template/scripts/git-safety-backup.sh --dry-run`. *You should see:* `WOULD WRITE` for each repository holding work GitHub lacks, and nothing written. It doubles as a list of unpushed work.
3. **Write, sealed:** `sh ~/code/my-template/scripts/git-safety-backup.sh --encrypt`. *You should see:* `WROTE` lines and `.bundle.enc` files. Once sealing is on, keep it on.
4. **Schedule it,** every two hours, from `scripts/launchd/git-safety-backup.plist.template`, after reading a dry run. The walkthrough has the lines.
5. **Get work back:** copy the bundle out of iCloud, open it with `--decrypt FILE`, clone from GitHub, then `git bundle verify /path/to/copy.bundle` and `git fetch /path/to/copy.bundle 'refs/*:refs/restored/*'`. *You should see:* `... is okay`, then the restored refs under `refs/restored/`, beside your own, so nothing collides.

5. Databases are readings, raw logs are the record

Why. A database built from logs answers questions quickly, but it holds nothing the logs do not. Back up what cannot be made again; rebuild what can.

- **The raw logs are the record.** Append-only files (JSONL and the like) are what a job wrote at the time. Keep them in git and push them while they fit. Never edit an old line; a correction is a new line.
- **The database is a reading.** Keep it gitignored (`*.sqlite`), out of the bundles, and out of Time Machine if it is large. Its loader rebuilds it from the logs, and a check proves the two agree. Restore means *rebuild*, not *copy back*.
- **When logs leave git.** Logs that outgrow a repository leave it on purpose, never by accident, and only after an off-machine copy exists and a restore from it has been rehearsed: files frozen per day, packed per month, snapshots to a drive and to an off-site bucket. The desk's database SOP holds one worked plan, with its measurements and a restore rehearsal.
- **Gitignored logs** (ticks, snapshots) are on this Mac only until Time Machine has a destination. Say so where they are written.

6. What never goes to GitHub

Never pushed	Why	Where it lives instead
Transcripts (`~/.claude/projects`) and their archives	they are private material: what people said, paths, pasted text, sometimes a token. On Nov 6, 2026 the lab's folder was 2.8 GB, with 13 files over 50 MiB and 2 over 100 MiB. GitHub warns above 50 MiB and **refuses any file over 100 MiB**, so one push would fail, or leave history that cannot be taken back	the Mac, the sealed archive, Time Machine
Keys, tokens, `.env` files, wallets	anything pushed is copied at once to every clone and cache. GitHub's push protection blocks many known secret formats, but not every one, and never a passphrase. If it blocks a push, do not bypass it: remove the secret, and rotate it if it ever left the Mac	the password manager
Passphrases (the archive's, the bundles')	without them the copies are noise; with them in a repository, the copies are open	the password manager, and one file on the Mac (mode 600)
Databases and their files	readings, rebuilt in seconds; binary, so every change is a whole new copy in history	rebuilt from the logs
Archives and bundles (`.enc`, `.bundle`)	large and binary; the public-lane tripwire refuses unread binaries	iCloud Drive or your `--dest`

Make it hard to slip: `.gitignore` the patterns (`.env` , `*.sqlite` , `*.enc` , `*.bundle`), keep secrets out of repository folders altogether, and let the tripwires of SOP-IT-004 speak. **Passphrases and keys live in the password manager,** one item each, named for what they open. Nobody reads one aloud to a helper, and no helper asks for one.

7. The restore drills

A drill restores into a new folder, never over live data. The person at the keyboard removes the restored folder afterwards if they want the space back; a helper leaves it and says where it is.

When	Drill	Command	Right looks like
Monthly, first week	transcripts still kept	the one-key check in part 1	`3650`
Monthly, first week	the newest archive opens	`claude-projects-archive.sh --list`, then `--verify FILE` on the newest	the list grows night by night; `verify` passes, nothing written
Quarterly	a full transcript restore	`--restore-latest ~/claude-restore-YYYYMM`, then `diff -rq` against `~/.claude/projects`	only files changed since the last archive differ
Quarterly	a git bundle	open one, then `git bundle verify` and `git fetch` into a fresh clone, as in part 4	`is okay`; the work under `refs/restored/`
Quarterly	one file from Time Machine	Finder › Browse Time Machine Backups (it mounts the backup: say so first), restore one file to a new place	the file opens, from a dated backup
Twice a year	the second key works	`--verify` an archive with `CLAUDE_ARCHIVE_PASSFILE` pointing at a file pasted from the password manager, not the Mac's own	`verify` passes: the password manager's copy opens the archives

Monthly	the space	`--list` total against the iCloud plan	under the plan with room to spare; otherwise `--move-old 3 DIR` to the drive

Drill log. One row per drill, newest last. A missed month is written down too.

Date	Drill	Result	By
2026-11-06	archive measured before first use: a full and a one-day incremental of the real folder, into scratch, with a scratch passphrase, then verified	full 1,913 files, 1.2 GB sealed, verified in 7 s; incremental 285 files, 216 MB. Not yet run into its real destination	a helper

Verification

The procedure is in place when all of these are true:

- the one-key check prints `3650`;
- `--list` shows an archive from the last night or two, and `launchctl print gui/$(id -u)/com.drasticstatic.claude-projects-archive` shows the agent loaded;
- both passphrases are in the password manager, and the clipboard was cleared;
- `git-safety-backup.sh --dry-run` lists nothing you did not expect, or bundles are being written;
- `tmutil latestbackup` gives a recent date, or the page says plainly that Time Machine has no destination yet;
- the drill log has a row from this quarter.

Troubleshooting

Symptom	Likely cause	What to do
`/resume` no longer lists a session from last month	`cleanupPeriodDays` is not set, or a project's own settings file sets it lower	run the one-key check and the `find` line in part 1; restore the session from the archive
The archive log says *Operation not permitted*, or `--status` says STALE	macOS keeps the background job out of iCloud Drive	schedule a folder outside iCloud (`--install --dest ~/Archives/claude-projects`) and make the iCloud copy by hand from Terminal weekly; never give `/bin/sh` Full Disk Access
refused: no passphrase file	`--init` never ran, or this is a new Mac	run `--init`, or put the file back from the password manager (the walkthrough's three lines)
refused: ... is mode 644; it must be 600	the passphrase file is readable by others	`chmod 600 ~/.config/claude-archive/passphrase`
refused: --dest ... does not exist	the drive is unplugged	plug it in; the next night tries again
`--list` shows a `.partial`	a run was cut off	nothing; the next run writes a whole archive
A restore says a file is missing in iCloud	Optimize Mac Storage moved it off the disk	open the folder in Finder and choose *Download Now*
A push is refused: *file exceeds GitHub's file size limit of 100 MiB*	a transcript, database or archive was staged	unstage it (`git restore --staged FILE`), add its pattern to `.gitignore`; if it is already in a commit that is not pushed, ask a steward before rewriting anything

A push is blocked by push protection	a secret is in a commit	do not bypass; remove it from the commit, and rotate the secret if it ever left the Mac
`tmutil destinationinfo` says there are no destinations	Time Machine is not set up	part 3, step 2; until then, say on the page that gitignored files have no copy

Related

SOP-IT-004 (the tripwires), SOP-IT-006 (a rename strands sessions; this page keeps them), SOP-IT-007, SOP-IT-008, SOP-IT-009, proposal 019 (*Backed up three ways*), the desk's database SOP (a worked off-machine copy for logs). In `my-template`: `walkthroughs/CLAUDE-PROJECTS-ARCHIVE.md`, `walkthroughs/ICLOUD-AND-GITHUB-TOGETHER.md`, `walkthroughs/CLAUDE-SESSION-ORDER.md`, `scripts/claude-projects-archive.sh`, `scripts/git-safety-backup.sh`, `scripts/install-claude-session-order.sh`.

Read on 2026-11-06: Claude Code's settings reference, `cleanupPeriodDays` (default 30, minimum 1, the sweep deletes without a message). Claude Code and GitHub may change these numbers; re-check them if this page ever seems wrong.

Change log

Date	Change
2026-11-06	A failed night is never silent: `--status`, the failure record, the banner and the `notify` hook; the morning-after check; what to do when macOS blocks the background job from iCloud Drive.
2026-11-06	Later the same day: an exclusion can be set before a drive exists, and how to take it off; hiding a folder from Finder (`chflags hidden`), and why it is neither privacy nor an exclusion; each destination gets its own schedule; `--move-old` and the three-months-in-iCloud rhythm; check the iCloud plan before scheduling; a renamed project folder is caught.
2026-11-06	Drafted, on the ask to bake transcript retention into how we protect, back up and store our data: retention at 3650 days, the nightly sealed archive and its measured sizes, Time Machine's exclusions, the safety bundles, readings and records, what never goes to GitHub, and a restore drill schedule with its log.

Part of the fleet's procedures. The master copies and the legend live in my-template.